

INFECTION CONTROL

NCLEX STUDY GUIDE

Asepsis • Isolation • PPE • Communicable Diseases

1. Chain of Infection and Asepsis

Link	Meaning / nursing control
Agent	Bacterium, virus, fungus or other pathogen.
Reservoir	Place where pathogen survives/grows.
Portal of exit	Cough, sneeze, saliva, blood, drainage or excreta.
Transmission	Contact, droplet, airborne or contaminated vehicle; nursing precautions interrupt this link.
Portal of entry	Eyes, nose, mouth, broken skin or invasive device.
Susceptible host	Person at risk because of age, illness, immunity or treatment.

Medical asepsis	Surgical asepsis
Clean technique: hand hygiene, clean gloves/PPE; hands rinsed downward. Soap/water when visibly dirty and always for C. difficile.	Sterile technique: all field items sterile; outer 1 inch contaminated; wet field contaminated; keep field above waist/in view; open far flap first; never turn back.

2. Standard Precautions, PPE and Sharps

- Use for every patient and all blood/body fluids except sweat, nonintact skin and mucosa. Perform hand hygiene before and after contact.
- Don PPE: gown, mask/respirator, goggles/face shield, gloves. Remove the most contaminated first; perform hand hygiene.
- N95 respirator for airborne rooms; surgical mask for droplet exposure and for isolated patients during transport.
- Never recap a needle unless unavoidable; then use one-hand scoop. Dispose in puncture-resistant container before three-quarters full. Clean spills with approved disinfectant/bleach dilution per policy.

C. DIFFICILE Use contact precautions and wash hands with soap and water; alcohol sanitizer does not reliably remove spores.

3. Transmission-Based Precautions

Contact	Droplet	Airborne
Gown + gloves, dedicated equipment, door may remain open. Examples: diarrheal illness/C. diff, RSV, MRSA/VRSA, impetigo, conjunctivitis, scabies, lice and localized zoster.	Surgical mask near patient; private/cohort room, door open; patient wears mask for transport. Examples: influenza, pertussis, diphtheria, rubella, mumps, bacterial meningitis and pneumococcal disease.	Negative-pressure room, door closed, N95 and 6-12 air exchanges/hr. Examples: TB, measles, varicella/smallpox.

MULTIPLE PRECAUTIONS Disseminated herpes zoster: airborne + contact + standard. Follow current facility/CDC guidance for emerging pathogens such as MERS, SARS and viral hemorrhagic fever.

4. Vector-Borne and Fungal Infections



Lyme disease: erythema migrans or bull's-eye lesion

- Lyme disease: deer-tick *Borrelia* infection. Remove tick with tweezers without squeezing; save/seal if required. Early flu-like illness and bull's-eye rash; later Bell palsy/cardiac findings and chronic arthritis/fatigue. Treat promptly with doxycycline/other antibiotic as ordered.
- Rocky Mountain spotted fever: tick exposure with fever/headache/myalgia and rash on wrists/ankles, palms/soles; use protective clothing, repellent and yard/tick control.



Tinea corporis (ringworm): fungal annular lesion

- Tinea corporis/pedis is fungal, not bacterial. Keep skin/feet dry and use antifungal such as clotrimazole, miconazole or ketoconazole; monitor liver toxicity with systemic therapy.
- Infectious mononucleosis: Epstein-Barr virus, fatigue, sore throat, lymphadenopathy and splenic enlargement. Standard precautions; avoid sharing utensils/kissing and contact sports. Left-shoulder pain may indicate splenic rupture.

5. Contact Diseases



Impetigo: characteristic honey-colored crust around the mouth

Disease	Key care
Impetigo	Contact until 24-48 hr after effective antibiotics; gentle cleansing/warm compress, topical antibiotic, separate linens. Watch edema or new murmur for post-strep complications.
Pediculosis	Permethrin and nit comb; boil/discard combs, hot-wash items, seal nonwashables about 2 weeks and vacuum. Nits alone do not indicate active transmission.
Scabies	Nocturnal pruritus and burrows in finger webs/wrists/axilla/groin; permethrin overnight, hot wash/dry, seal items and repeat treatment. Lindane contraindicated in young children/pregnancy.

6. Droplet Diseases

Disease	Hallmark / priority
Rubella	Pink rash face-to-body and posterior/occipital nodes; contagious 7 days before/after rash. Avoid pregnant contacts. Postpartum MMR when nonimmune; avoid pregnancy afterward per current guidance.
Pertussis	Catarrhal stage then paroxysmal whoop/vomiting. Droplet until effective antibiotic period; humidified oxygen/suction and reduce irritants.
Diphtheria	Gray pseudomembrane, foul discharge and “bull neck”; airway equipment, antitoxin and antibiotics.
Scarlet fever	Strep: strawberry tongue, sandpaper rash and Pastia lines. Watch glomerulonephritis or rheumatic fever.
Mumps	Parotid pain/swelling, worse with sour foods; orchitis/infertility and meningitis risk. Scrotal support and compresses.
Parvovirus B19	Slapped-cheek rash; no longer contagious once rash appears. Avoid pregnant contacts.

7. Airborne Diseases

Disease	Key findings / care
Measles (rubeola)	3 Cs: cough, coryza, conjunctivitis; Koplik spots and descending rash. Airborne; cool mist, dim light for photophobia and no aspirin.
Varicella	Lesions in multiple stages simultaneously; airborne + contact until

Disease	Key findings / care
Herpes zoster	all crusted. Calamine/oatmeal, antihistamine, mittens/trim nails and no aspirin. Dermatomal burning/allodynia; localized contact, disseminated airborne + contact. Moist dressings, early antiviral and neuropathic pain drug; urgent eye evaluation for facial/ocular disease.
Tuberculosis	Airborne and N95 for all entering. Continue isolation until criteria met per public-health/facility policy.

8. Emerging and High-Consequence Infections

- MERS/SARS: severe coronavirus respiratory disease with fever, cough, dyspnea and respiratory failure; use current enhanced precautions.
- Ebola/viral hemorrhagic fever: fever, petechiae and hemorrhage; strict PPE, contact/droplet and added airborne protection for aerosol procedures per protocol.

9. Biological and Chemical Agents

Agent	Hallmark / response
Anthrax	Not usually person-to-person. Inhaled: severe respiratory illness/hemoptysis; cutaneous: painless black eschar; GI: bleeding. Decontaminate and treat with ciprofloxacin/doxycycline.
Botulism	Home-canned food/honey in infant; descending paralysis, ptosis, dysphagia and respiratory failure. Airway and antitoxin.
Plague	Bubonic nodes, septicemia or pneumonic cough; pneumonic form requires droplet. Streptomycin/gentamicin.
Smallpox	High fever and dense facial rash; airborne/contact and public-health emergency.
Tularemia	Rabbit/tick exposure, fever, weakness and pneumonia; aminoglycoside/doxycycline.
Mustard gas	Pain-free blisters, eye injury and respiratory burns; decontaminate/supportive care.
Sarin	Cholinergic SLUDGE findings, seizure and paralysis; atropine + pralidoxime.

10. Immunization Review

1	2	4	6	12 15 18	19-23 2 4-6 11-12 16
Birth	M	onths			Years
Hep B #1	Hep B #2			Hep B #3	
RV #1	RV #2	RV* #3			
DTaP #1	DTaP #2	DTaP #3			
DTaP #1	DTaP #2	DTaP #3	DTaP #4		DTaP #5

Vaccination schedule - birth through early infancy

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Hib	Hib	Hib	Hib	
#1	#2	#3	#4	
PCV	PCV	PCV	PCV	
#1	#2	#3	#4	
IPV	IPV		IPV	IPV
#1	#2		#3	#4
			Hep A	
			#1&3	
			MMR	MMR

Vaccination schedule - later infancy and childhood

			#1	#2
			VZV	VZV
			#1	#2
				Tdap
				#1
				HPV**
				#1,2
				MCV MCV
				#1 #2

Vaccination schedule - childhood and adolescent vaccines

Vaccination Abbreviations

Abbreviation	Meaning
Hep B	Hepatitis B vaccine
RV	Rotavirus vaccine: 2 doses when Rotarix is used; 3 doses when RotaTeq is used
DTaP	Diphtheria and tetanus toxoids with acellular pertussis vaccine
Tdap	Tetanus toxoid, reduced diphtheria toxoid and acellular pertussis vaccine
Hib	Haemophilus influenzae type b conjugate vaccine
PCV	Pneumococcal conjugate vaccine
MenB	Serogroup B meningococcal vaccine
IPV	Inactivated poliovirus vaccine
MMR	Measles, mumps and rubella vaccine
VZV	Varicella-zoster virus vaccine
Hep A	Hepatitis A vaccine
MCV	Meningococcal conjugate vaccine
HPV	Human papillomavirus vaccine; 3-dose series when initiated at age 15 or older

- Verify current CDC schedule, contraindications, pregnancy status and immune suppression before live vaccines.
- Vaccines for hepatitis A and B prevent those infections; hepatitis C/D/E do not have routine U.S. vaccines (HBV vaccination prevents hepatitis D coinfection).
- Document vaccine, lot, site, route, date and education; observe and manage allergic reactions.

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FINAL NCLEX FOCUS Select isolation by route, protect the airway first, perform correct hand hygiene/PPE, prevent sharps injury and report neurologic, respiratory or hemorrhagic deterioration immediately.